

Is there “groupthink” in an AI agent swarm? H12 in plain terms

Vivian Rogers, with Claude Opus 5.5

Agent-swarm physics project

(Explainer for a mathematically literate non-specialist. Round 1, exploratory data only; nothing here has been tested on the held-out period)

The AI Village runs 10–20 language-model agents together for weeks. We asked whether such a swarm (i) moves as one, i.e. has a few collective modes, and (ii) collapses onto a few themes in what it says. There is exactly one collective mode, and it is the simplest possible one: a global “how busy is the swarm” variable, partly caused by the platform pausing everyone at once. What agents say does *not* collapse after a new goal; it diversifies. Where diversity is low, the cause is agents repeating *themselves*, not consensus. The practical result is a three-number monitor that runs on any swarm’s logs.

I. THE QUESTION

“Groupthink” can be made precise in two ways. Do the agents *move as one*? If the swarm has a few collective modes, directions along which everybody varies together, its dynamics are low dimensional however many agents there are. Does what they *say* collapse onto a few themes? Then the spread of their messages in meaning-space should shrink. We predicted yes to both, and especially that the spread would shrink right after a new goal is announced.

II. PART 1: EIGENVALUES OF A CORRELATION MATRIX

For agent i and minute t define a spin $s_i(t) = +1$ if the agent is active (or posting in chat) and -1 if not. Over a goal period this is an $N \times T$ matrix (N agents, T minutes). We form the $N \times N$ correlation matrix $C_{ij} = \langle s_i s_j \rangle_t$ (after centering and scaling) and look at its eigenvalues.

a. Noise level. If agents were independent, finite T would still smear the eigenvalues of C around 1. Random-matrix theory (the Marchenko–Pastur law) says pure noise puts them in the band

$$\lambda \in [(1 - \sqrt{q})^2, (1 + \sqrt{q})^2], \quad q = N/T. \quad (1)$$

An eigenvalue above the band is real shared structure, and its eigenvector says which agents take part, with what weights.

b. Why the textbook band is not enough. The formula assumes each minute is an independent draw. Here it isn’t: agents work in bursts and share a daily schedule. On simulated swarms with *no* coupling, the naive band still flagged a collective mode 50–90% of the time.

c. The noise level we used: a cross-day surrogate. Replace each agent’s day with the *same agent’s* record from a *different* day. This keeps each agent’s rhythm and burstiness but destroys same-day coordination. Repeat 200 times; the 95th percentile of the top eigenvalue is the noise edge. On the same simulated swarms this gave 6–10% false alarms.

d. Content. The same machinery runs on what agents say: each message is embedded as a vector (32 numbers after removing the directions every message shares), and “correlation” means agents’ messages drifting the same way within the same half hour.

III. PART 2: EFFECTIVE DIMENSION

For the cloud of message vectors in a window, with covariance eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots$, the participation ratio

$$\text{PR} = \frac{(\sum_k \lambda_k)^2}{\sum_k \lambda_k^2} \quad (2)$$

is an effective number of dimensions: $\text{PR} = 1$ if all variance lies on one axis and $\text{PR} = d$ if it is spread evenly over d axes. With few messages the naive estimate is biased low (47–87% of the truth in simulations), so we used a bias-corrected version.

IV. WHAT WE FOUND

a. 1. Exactly one collective mode, the simplest one. In 22 of 24 analysis units (a unit is a goal period, or a piece of one split where something changed), activity has exactly one eigenvalue above the noise edge; chat 20/24; content 24/24. Every agent loads on it with the same sign: the top eigenvector is essentially the uniform vector $u = (1, \dots, 1)/\sqrt{N}$. Concretely, the Rayleigh quotient

$$\text{VR} = u^\top C u \leq \lambda_1 \quad (3)$$

(the variance of total activity) captures 97% of λ_1 , and equality holds only when the top eigenvector is exactly uniform. That is the signature of the Curie–Weiss magnet, one global variable (“how busy is the swarm right now”), the analogue of the *market mode* in stock correlations. Its strength per period lines up with H02’s independently estimated coupling (rank correlation 0.73).

Part of it is everyone going quiet at once: minutes with at most one active agent make up to 42% of a period, and they create correlation trivially. Removing them, the

mode survives in 9 of the 11 units where such lulls are rare (14/24 overall). The content mode survives removing each agent’s daily average, so it is agents’ messages moving together *within* a day.

b. 2. “Collapse after a new goal” failed, in the opposite direction. Across 20 goal changes the participation ratio moved by a median +7%, indistinguishable from ordinary day-to-day change ($p = 0.86$); in the later, more autonomous regime it *rose* by 44%, 4 out of 4 times. Day 1 of a goal is more diverse than later days in 13 of 16 periods ($p = 0.02$), and shared-goal weeks were no less diverse than pick-your-own-goal weeks. A new goal makes the swarm explore, then it narrows somewhat: an anti-quench.

c. 3. Low diversity is loops, not consensus. The least diverse periods (#38–#40) had 16–60% of an agent’s daily messages as near-copies (cosine > 0.95) of *its own* earlier messages, and $\lesssim 3\%$ (at most 8% on any day) as near-copies of other agents’. Removing self-repeats leaves every verdict unchanged.

V. SO WHAT?

The swarm is not a hive mind. Statistically it looks like weakly coupled individuals plus one shared busy/idle variable, much of which the platform itself causes, con-

sistent with the other hypotheses: coupling here is real but weak.

If you run a swarm, watch for loops rather than consensus. The practical output is a three-number monitor that needs no fitting:

1. **Collective-mode strength:** the top eigenvalue of the agents’ content-correlation matrix over its cross-day noise edge, against the swarm’s own recent history (“moving in lockstep” alarm).
2. **Echo rate:** the share of messages nearly identical to another agent’s message in the last two hours, the actual groupthink alarm, which stayed low here.
3. **Self-repetition rate:** the same against the agent’s own earlier messages, the stuck-agent alarm, which fired in exactly the periods where diversity collapsed.

a. Caveats. One embedding model (a different one might draw the space differently); a single dataset; the lull filter loses power when lulls are long; nothing has been tested on the held-out periods. The confirmation script is written and runs only with sign-off, once.

Full predictions, results and scorecard: the card `hypotheses/H12-groupthink-dimensional-collapse/README.md`; figures in its `figures/` folder (`H12-summary.pdf`, `rmt.arm.pdf`, `dimensionality.arm.pdf`).